



# SOC ANALYST PROGRAM

*Learn Security. Build Detection Skills. Defend Real-World Systems.*

SKILL IT Education presents a structured, practical career pathway into Security Operations — taking learners step-by-step from IT and networking foundations to SIEM monitoring, incident response and real-time threat detection. Every concept is reinforced through hands-on labs, real-world tools and industry-aligned SOC practices.

Cyber threats now target every organization, and businesses across every industry depend on skilled SOC analysts to monitor, detect and respond before damage is done — making this one of the fastest-growing and most valuable career paths in technology today.

## PROGRAM DURATION

|                     |          |
|---------------------|----------|
| Total Duration      | 7 Months |
| Structured Learning | 5 Months |
| Industry Internship | 2 Months |

## GET IN TOUCH

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# Program Overview

*Build Your SOC Analyst Career*

This program takes you step-by-step from strong IT and networking foundations to real Security Operations Center work — monitoring networks, operating SIEM platforms, investigating incidents and hunting threats in live, hands-on environments.

## WHY THIS MATTERS

Cyber attacks now target every industry, and organizations urgently need engineers who can monitor, detect and respond — not just install security tools. SOC skills are among the highest in demand across the global technology job market.

*From Learning to Monitoring. From Alerts to Action. From Skills to Career.*

1

**Foundations: IT, Networking & Operating Systems**

2

**The Cyber Threat Landscape**

3

**Inside a Security Operations Center**

4

**SIEM Platforms & Security Monitoring**

5

**Incident Response & Threat Hunting**

# Module 1 — Foundations: IT, Networking & OS

## WHAT YOU WILL LEARN

- IT infrastructure & software fundamentals
- Networking essentials: LAN, MAN, WAN & Internet
- Network topologies, devices & the OSI/TCP-IP models
- IP addressing, classifications & subnetting
- Routers, switches, firewalls & IDS/IPS placement
- Windows & Linux installation and administration
- Linux terminal / CLI & command-line practice
- Reputation checks for suspicious IPs & URLs

## TOOLS & TECHNOLOGIES

Windows Server

Linux / Ubuntu

CLI

Wireshark

Nmap

VirtualBox

## NETWORKING STACK

IP

Subnet

Router

Switch

Firewall

IDS/IPS



**Learning Outcome:** Students build a strong technical foundation for security work by configuring networks, working across Windows and Linux, and understanding how IT infrastructure actually operates.

# Module 2 — The Cyber Threat Landscape

## WHAT YOU WILL LEARN

- Threat, vulnerability & risk — how they connect
- CIA Triad and AAA security models
- Ethical hacking vs. hacking, and types of hackers
- The five phases of hacking: recon to clearing tracks
- Malware types: viruses, worms, trojans & ransomware
- Phishing, spear-phishing & business email compromise
- Password attacks, MITM, DoS & DDoS attacks
- Web application attacks: injection, XSS & more

## COMMON ATTACK TYPES

Phishing

Ransomware

DDoS

MITM

SQL Injection

XSS

Brute Force

Social Eng.

## DEFENSIVE MEASURES

Patch Mgmt

Hardening

DLP

HIPS/NIPS



**Learning Outcome:** Students learn to recognize, classify and understand the mechanics of common cyber attacks, building the attacker's-eye view every SOC analyst needs.

# 05 Module 3 — Inside a Security Operations Center

## WHAT YOU WILL LEARN

- What a SOC does, and why it matters
- Key SOC functions: triage, investigation & hunting
- How a SOC is structured and staffed
- Key SOC roles and responsibilities
- Modern-day SOC: people, process & technology
- SOC as a Service (SOCaaS) and delivery models
- NOC vs. SOC — how the two teams differ
- Cyber security monitoring essentials

## TOOLS & TECHNOLOGIES

|           |             |      |
|-----------|-------------|------|
| SIEM      | NIDS / NIPS | SOAR |
| EDR / XDR | DLP         | IAM  |
| Firewalls | Vuln. Mgmt  |      |

## SOC TIER STRUCTURE

|                |                       |                        |
|----------------|-----------------------|------------------------|
| Tier 1: Triage | Tier 2: Investigation | Tier 3: Threat Hunting |
|----------------|-----------------------|------------------------|



**Learning Outcome:** Students understand how a SOC is structured and staffed, and can explain the tools, roles and workflows that keep a security team running around the clock.

## 06 Module 4 — SIEM Platforms & Security Monitoring

### WHAT YOU WILL LEARN

- SIEM architecture, event & flow collectors
- IBM QRadar: dashboards & offense analysis
- Malicious IP communication & phishing analysis
- Windows, Linux & security device log types
- Correlation rules, false positives & alert triaging
- Reference sets, use-case creation & rule tuning
- Log source integration across the enterprise
- Generating governance & security reports

### TOOLS & TECHNOLOGIES

IBM QRadar

Splunk

Win Collect

Firewall Logs

IPS / WAF

Proxy Logs

### SIEM WORKFLOW

Collect

Correlate

Detect

Investigate

Report



**Learning Outcome:** Students learn to operate a SIEM platform end to end — building dashboards, investigating offenses, tuning rules and producing real security reports.

# 07 Module 5 — Incident Response & Threat Hunting

## WHAT YOU WILL LEARN

- Incident handling process & response lifecycle
- Preparation, containment, eradication & recovery
- Security incident playbooks vs. runbooks
- The Cyber Kill Chain and MITRE ATT&CK framework
- Cyber Threat Intelligence (CTI) and its lifecycle
- Indicators of Compromise (IOC) vs. Indicators of Attack
- Threat hunting methods & investigation tools
- Email header analysis & malware analysis basics

## FRAMEWORKS & TOOLS

MITRE ATT&CK

Kill Chain

CyberChef

Sysinternals

Google Dorks

IOC / IOA

## INCIDENT RESPONSE LIFECYCLE

Prep

Identify

Contain

Eradicate

Recover

Review



**Learning Outcome:** Students learn to respond to real incidents using a structured lifecycle, and to hunt proactively using threat intelligence and industry-standard frameworks.



# SOC Technology Stack

*The complete toolset used across every module of the program*

## Networking

- Routers & Switches
- Firewalls
- IDS / IPS

## Operating Systems

- Windows Server
- Linux / Ubuntu
- CLI Tools

## SIEM Platforms

- IBM QRadar
- Splunk
- Win Collect Agent

## Security Monitoring

- SOAR
- EDR / XDR
- DLP & IAM

## Threat Intelligence

- MITRE ATT&CK
- Cyber Kill Chain
- IOC / IOA Feeds

## Investigation Tools

- CyberChef
- Sysinternals
- Google Dorks

09

# Hands-On Labs & Project-Based Learning

*Build What You Learn — every module is reinforced with hands-on work*

01

## Network & Lab Environment Setup

Configure IP addressing, subnetting and a virtual lab with Windows and Linux systems.

02

## SIEM Monitoring Lab

Onboard log sources into IBM QRadar, create reference sets and build real-time dashboards.

03

## Threat Detection Exercise

Analyze malware, phishing emails and malicious IP activity using live SIEM offense data.

04

## Incident Response Simulation

Run through a full incident lifecycle — from identification to containment and recovery.

05

## Threat Hunting Project

Use MITRE ATT&CK and threat intelligence feeds to proactively hunt for hidden threats.

**FINAL CAPSTONE**

## End-to-End SOC Simulation

Monitor, investigate and respond to a simulated multi-stage attack from detection to resolution.

# Learning Outcomes

*What You Will Be Able to Do by the End of the Program*

- ✓ Configure and troubleshoot enterprise networks
- ✓ Administer Windows Server and Linux environments
- ✓ Operate SIEM platforms such as IBM QRadar & Splunk
- ✓ Read and correlate security device logs
- ✓ Recognize and classify common cyber attacks
- ✓ Perform basic malware & email analysis
- ✓ Monitor, triage and escalate security alerts
- ✓ Apply the Cyber Kill Chain and MITRE ATT&CK
- ✓ Follow a structured incident response lifecycle
- ✓ Use threat intelligence to strengthen detection
- ✓ Conduct proactive threat hunting
- ✓ Communicate findings through reports & playbooks

# Career Opportunities

*Build Your Career in Security Operations*

## SOC Operations

- SOC Analyst (L1)
- SOC Analyst (L2)
- Security Analyst

## Incident & Threat Response

- Incident Response Analyst
- Threat Intelligence Analyst
- Junior Threat Hunter

## Security Engineering

- SIEM / Security Engineer
- Vulnerability Analyst
- Network Security Engineer

## Monitoring & Detection

- Security Monitoring Analyst
- Alert Triage Specialist
- Log Analysis Engineer

## Compliance & Risk

- Cyber Risk Analyst
- IT Security Auditor
- Compliance Analyst

## Advanced Career Paths

- SOC Team Lead
- Security Architect
- CISO (long-term path)

*Career outcomes depend on individual skills, project execution, interview performance, experience level and market requirements.*



WHY CHOOSE SKILL IT EDUCATION

# DON'T JUST LEARN SECURITY. LEARN TO DEFEND IT.

*Learn. Monitor. Detect. Respond. Defend.*

SKILL IT Education builds SOC analysts, not just certificate holders. Through an industry-driven curriculum, hands-on labs and real SIEM tools, every student graduates with practical monitoring skills, verified experience and the confidence to walk into a security operations role from day one.

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*Let's build a safer future.*

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